

Chapter 39: Sterilization — Revision Notes

Williams Obstetrics 26e, pp. 1759–1776 of the PDF. High-yield numbers are in **bold**. This chapter has no tables. All 5 figures are included from the PDF.

Big picture: **28%** of US women using contraception rely on male or female sterilization. It is for women who **clearly understand its permanence** (reversal is difficult and often unsuccessful); alternatives must be offered, and after counseling **patient autonomy is respected**. Puerperal sterilization follows **~7% of US live births**. **Salpingectomy** is now recommended for consideration because it also lowers ovarian cancer risk. **Vasectomy is safer and more effective** than tubal sterilization.

1. Puerperal Tubal Sterilization

Definitions

- **Puerperal** sterilization = done after cesarean or vaginal delivery.
- **Nonpuerperal (interval)** sterilization = done at a time unrelated to recent pregnancy.
- Methods: **occlusion, excision or division** of the fallopian tubes.

Timing

- Several days postpartum the **fundus is at the umbilicus** and the tubes lie directly beneath the abdominal wall; abdominal laxity lets the incision be moved over each cornu.
- Parkland practice: surgery the **morning after delivery** by a dedicated team.
 - Minimizes hospital stay, lowers the chance that **postpartum hemorrhage** complicates recovery, and lets the **newborn's status** be confirmed first.
- Alternative: operate **immediately after delivery**, using the labor **neuraxial catheter**.
- Designating postpartum sterilization as **nonelective** lessens access barriers (busy units prioritize OR time for intrapartum cases).
- Postpartum tubal ligation is safe **regardless of BMI**.

Method selection

- Standard: excise a **midtubal segment**; the cut ends seal by **fibrosis and peritoneal regrowth**.
- Common methods: **Parkland, Pomeroy, modified Pomeroy**.
- **Filshie clips**: used less often, with **slightly lower efficacy**.
- **Hysterectomy solely for sterilization** is hard to justify: much higher surgical morbidity.

Salpingectomy and ovarian cancer

- Most **pelvic serous cancers arise from the distal fallopian tube**.
- **SGO and ACOG** recommend considering salpingectomy. For women at **average risk** of ovarian cancer, discuss risk-reducing salpingectomy with abdominopelvic surgery, with hysterectomy, or **in lieu of tubal ligation**.

Point	Figure
Lifetime ovarian cancer risk	~1%
Tubal interruption alone	~30% ↓ ovarian cancer
Salpingectomy	42–78% ↓
Ovarian reserve (AMH)	No difference vs tubal ligation (small studies)
Extra operating time at cesarean	5–10 min longer than partial salpingectomy
Blood loss	Comparable or slightly higher; no ↑ transfusion or hematocrit fall

- Caveats: no adequately sized prospective studies in low-risk women; few data on ovarian reserve; only small studies after **vaginal delivery**.

Technique

Anesthesia and preparation

- **Spinal** analgesia for cases on postpartum day 1. **General anesthesia less desirable** (residual aspiration risk).
- Near delivery, reuse the **labor epidural catheter**.
- Thrombocytopenia: platelets **>70,000/μL** for spinal blockade.
- **Empty the bladder**: avoids laceration, and a full bladder pushes the fundus and tubes above the umbilicus.
- **Clean case** → **no antibiotic prophylaxis**.

Incision: infraumbilical

- Why infraumbilical:
 1. The fundus lies near the umbilicus.
 2. The umbilicus is the **thinnest** part of the anterior abdominal wall (less dissection to the linea alba).
 3. The fascia there is strong → low **incisional hernia** risk.
 4. Good cosmesis in the lower umbilical skin fold.
- Skin incision **2–3 cm** (transverse or vertical); **4–5 cm** in obese women.
- Bluntly separate subcutaneous fat (Allis clamp opened/closed, or two army-navy retractors pulling in opposite directions) so the fascia is clean for incision and closure.
- Fascial incision follows the skin orientation. Grasp the linea alba with **two Allis clamps**, one each side of the planned incision (Fig 39-1); the peritoneum is often entered at the same time. Extend with curved Mayo scissors if too small.

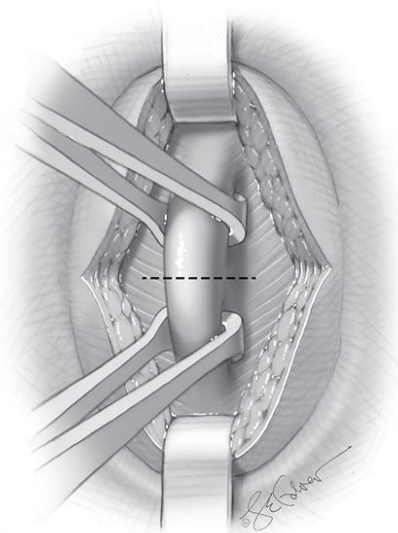


Figure 39-1. Linea alba fascia lifted between two Allis clamps away from the viscera; dashed line = planned fascial incision.

Exposure

- Army-navy or appendiceal retractors; narrower, deeper retractors and a larger incision if obese.

- **Trendelenburg** displaces bowel/omentum cephalad.
- A single moist fanned-out gauze can pack bowel, but **always clamp a hemostat to its end** (avoid retained sponge).
- Tilt the table away from the side being exposed.

Identifying the tube

- Grasp the tube at its **midportion with a Babcock**, then a second clamp more distally, and follow it until the **fimbriated end is seen**.
- ⚠ A common cause of failure is ligating the **wrong structure, usually the round ligament**.
- If the tube is dropped, the **identification must be repeated**.

Ligation methods

Method	Steps	Key point
Parkland	Perforate an avascular site of mesosalpinx with a small hemostat; open the jaws to free ~2.5 cm of tube. Ligate proximally and distally with 0-chromic ; excise the ~2-cm segment between; check hemostasis.	Designed to avoid the close contact of cut ends that occurs with Pomeroy
Pomeroy	Midsegment tubal loop tied at its base; loop excised.	Plain catgut → rapid absorption so the severed ends separate
Total salpingectomy	Mesosalpinx sequentially clamped, cut and ligated ; clamps across tube and mesosalpinx at the cornu before transection.	Needs a larger umbilical incision; risk of bleeding from large congested mesosalpingeal veins (may need extension or adnexectomy). Bipolar sealing devices (LigaSure, ENSEAL) are faster but cost more.

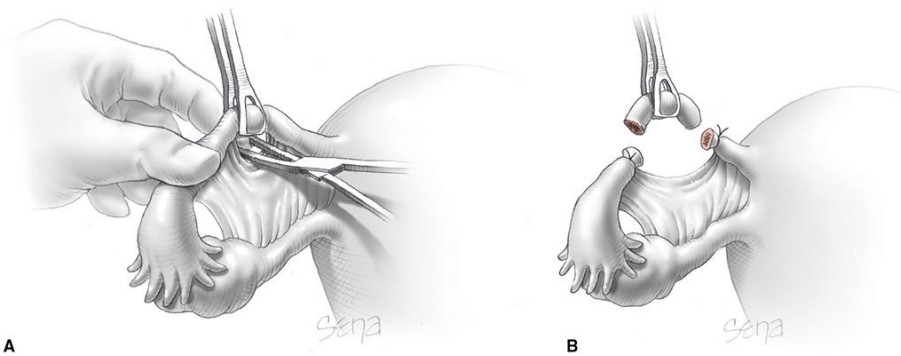


Figure 39-2. Parkland method: (A) hemostat opens an avascular window in the mesosalpinx; (B) ~2-cm segment excised between two 0-chromic ties.

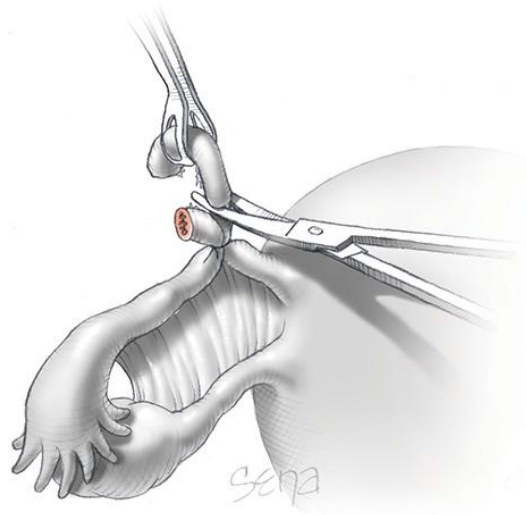


Figure 39-3. Pomeroy method: a midsegment loop is tied with plain catgut and excised; absorption lets the ends pull apart.

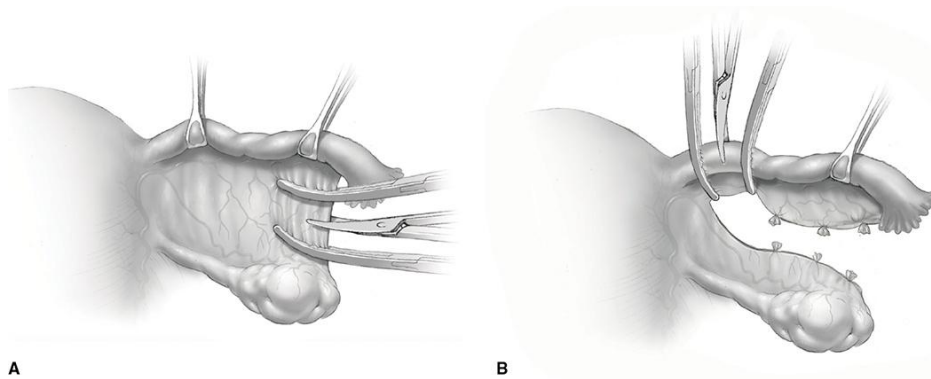


Figure 39-4. Salpingectomy: (A) mesosalpinx clamped, cut and ligated in sequence; (B) clamps placed at the cornu before transecting the tube.

Postoperative care

- Diet as tolerated; early walking.
- **Ileus is infrequent** → **suspect bowel injury** (rare).
- Most women go home on **postoperative day 1**.

2. Nonpuerperal (Interval) Tubal Sterilization

- Three approaches:
 1. **Ligation and resection** at laparotomy (as for puerperal).
 2. **Rings, clips or inserts** placed laparoscopically or hysteroscopically.
 3. **Electrocoagulation** of a tubal segment, usually laparoscopic.
- **Laparoscopy is the most common US approach**: ambulatory, usually **general anesthesia**, home within hours.
- **Minilaparotomy (3-cm suprapubic incision)**: popular in resource-poor countries.
- Major morbidity is **rare** with either.
- Rarely: **colpotomy** through the posterior vaginal fornix.

3. Long-Term Complications

Contraceptive failure (CREST study, 10,863 women, 1978–1986)

- **Cumulative failure 18.5 per 1000 (~0.5%)** across all tubal methods.
- Puerperal sterilization is highly effective: **5 per 1000 at 5 yr, 7 per 1000 at 12 yr**.
- **Two reasons puerperal sterilization fails:**
 1. **Surgical error:** transecting the **round ligament** or only part of the tube → **send both tubal segments for pathology**.
 2. **Fistula or spontaneous reanastomosis** between the stumps.
- **Ectopic pregnancy:** ~30% of pregnancies after failed tubal sterilization; **20%** after a postpartum procedure.

Pearl: any pregnancy symptoms after tubal sterilization → **exclude ectopic pregnancy**.

Other effects

- **Salpingitis** afterward is **highly unlikely**.
- Heavy or intermenstrual bleeding is **not increased** (most studies).
- Sexual interest/pleasure **unchanged in 80%**; of the 20% reporting change, **positive changes were 10–15x more likely**.
- **Regret:** more common when sterilized at a **younger age**.
 - **7%** of women regretted tubal ligation by **5 years** (CREST).
 - **6.1%** of women whose husbands had a vasectomy had similar regret.

Tubal sterilization reversal

- Neither reversal surgery nor ART guarantees later fertility: difficult, expensive, not always successful.
- **Better pregnancy rates with:** **≥7 cm of remaining tube**, age **<35 yr**, **short interval** since sterilization, **isthmic–isthmic** repair.

Reversal	Outcome
Reanastomosis via laparotomy	Live births 44–82%
Ectopic rate after reanastomosis	2–10%
Reversal of Essure	Live births 0–27%

4. Transcervical Sterilization

- Hysteroscopic device placed into the **proximal tubes** via the ostia. **No transcervical method is currently approved in the US**.
- **Essure** and **Adiana** have both been **withdrawn from the US market**.

Device	Description	Notes
Adiana	Cylindrical nonabsorbable silicone elastomer matrix	Stimulates tissue ingrowth → occlusion
Essure	Long, slender, coiled metallic insert	Fibroblastic proliferation → occlusion. Failure <1% to 5%

Essure adverse effects

- **Chronic pelvic pain in 2–6%:** from tubal perforation, device migration, or the device itself.
- Removal: **proximal linear salpingostomy**; ⚠ removal does **not** cure every symptomatic patient.
- Also abnormal bleeding and **allergy/hypersensitivity** to device components.

Chemical occlusion: quinacrine

- Pellets placed in the uterine fundus with an IUD-type inserter; inflammation occludes the tubes.
- Pregnancy rates **1% at 1 yr, 12% at 10 yr**.
- **WHO recommends against** it (carcinogenesis concerns). Some still consider it for resource-poor countries.

5. Vasectomy

- Up to **half a million US men** per year; **5% of women** rely on it.
- The **vas deferens lumen** is disrupted so sperm cannot pass from the testes.

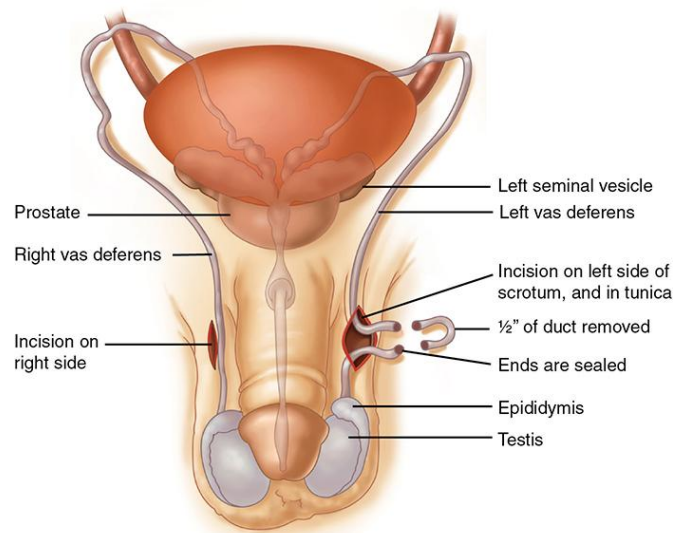


Figure 39-5. Vasectomy: scrotal incision on each side, about 1/2 inch of vas deferens removed, and the ends sealed.

Vasectomy vs tubal sterilization

- Vasectomy is **safer**: less invasive, done under **local analgesia**.
- Tubal sterilization has a **20-fold higher complication rate** and a **10- to 37-fold higher failure rate**.
- **⚠ Not immediately effective**: stored sperm take **~3 months or 20 ejaculations** to clear.
- **AUA: semen analysis at 8–16 weeks** to document sterility; use **other contraception until azoospermia** is confirmed.

Failure

- **9.4 per 1000** in the 1st year; **11.4 per 1000** cumulative at 2, 3 and 5 years.
- Causes: **intercourse too soon** after ligation, **incomplete occlusion**, **recanalization**.
- Reversal is best done **microsurgically**. Conception after reversal is worse with **longer time since vasectomy**, **poor sperm quality** at reversal, and the **type of reversal** needed.

Quick-Recall: Sterilization Methods Compared

Method	Setting	Failure / efficacy	Key risk or point
Parkland / Pomeroy (partial salpingectomy)	Puerperal, infraumbilical minilaparotomy	5/1000 at 5 yr, 7/1000 at 12 yr	Ligating the round ligament; ~20% of failures ectopic
Filshie clip	Puerperal or interval	Slightly less effective	—
Total salpingectomy	At cesarean, or instead of ligation	Sterilization + 42–78% ↓ ovarian cancer	5–10 min longer; mesosalpinx bleeding
Laparoscopic (rings, clips, coagulation)	Interval, ambulatory	CREST overall 18.5/1000	General anesthesia
Essure (withdrawn)	Hysteroscopic	<1–5%	Chronic pelvic pain 2–6%
Quinacrine	Resource-poor settings	1% (1 yr), 12% (10 yr)	WHO advises against (carcinogenesis)
Vasectomy	Office, local analgesia	9.4/1000 yr 1; 11.4/1000 at 5 yr	Not immediate: 3 mo / 20 ejaculations ; semen analysis 8–16 wk

Key Numbers to Memorize

- **28%** of US contraceptive users rely on sterilization; puerperal sterilization after **~7%** of live births; vasectomy relied on by **5%** of women
- Skin incision **2–3 cm** (obese **4–5 cm**); platelets **>70,000** for spinal; no antibiotic prophylaxis
- Parkland: free **2.5 cm**, excise **2 cm**, **0-chromic**; Pomeroy: **plain catgut**
- Ovarian cancer lifetime risk **~1%**; tubal ligation **↓30%**; salpingectomy **↓42–78%**
- CREST cumulative failure **18.5/1000 (~0.5%)**; puerperal **5/1000 (5 yr), 7/1000 (12 yr)**
- Ectopic: **30%** of failures overall, **20%** after postpartum procedures
- Regret **7%** at 5 yr; sexual function unchanged in **80%**
- Reversal: **≥7 cm** tube, age **<35**; live births **44–82%**; ectopic **2–10%**; Essure reversal **0–27%**
- Essure pelvic pain **2–6%**; Essure failure **<1–5%**
- Vasectomy: tubal sterilization has **20×** complications and **10–37×** failures; azoospermia after **3 mo / 20 ejaculations**; semen analysis at **8–16 wk**